
Final Report

Introduction to Machine Learning Safety CARLA Autonomous Driving Perception System

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GitHub Repository: <https://github.com/rushabhkayadra/carla-ml-safety.git>

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1. System Description

Provided system description

An autonomous vehicle (AV) prototype operates in an urban environment. The system is composed of the following components:

1. **Camera:** A single forward-facing RGB camera, rigidly mounted behind the windshield, provides images at 10 Hz. There are *no other perception sensors* (no radar, no LiDAR, no ultrasonic).
2. **Perception models:** Three separate binary classifiers each receive the camera frame and output one binary prediction: *pedestrian present*, *vehicle present*, or *traffic light present*. Each model was trained exclusively on sunny daytime data collected in the CARLA simulator.
3. **Distance oracle:** A separate model estimates the distance to detected pedestrians and vehicles. You may assume this model is perfect for the sake of this course.
4. **Planning module:** The rule-based planner consumes the three perception outputs together with the current vehicle speed and steering angle. Its primary safety-relevant decisions are whether to continue driving, request deceleration, or issue an emergency brake command. Emergency braking is triggered when the pedestrian detector reports a pedestrian within a critical distance, when the vehicle detector indicates a road user ahead within a speed-dependent threshold, or when the traffic-light detector indicates that stopping is required at an approaching intersection. The planner does not have access to raw camera images and relies entirely on the perception models' outputs.
5. **Vehicle actuators:** Throttle, brake, and steering commands are executed by drive-by-wire actuators. Emergency braking applies maximum deceleration within the physical limits of the vehicle.
6. **Human safety operator:** A trained operator sits in the driver seat and monitors the road. The operator can override the autopilot at any time by pressing the brake pedal or turning the steering wheel. The operator receives a visual dashboard showing the current perception output and system status, but no auditory alerts are provided. During testing, operators work 4-hour shifts.
7. **Operating environment:** The vehicle is tested on public urban roads at speeds up to 50 km/h. The intended conditions are daytime, dry weather, and mapped intersections. However, weather and lighting conditions can change during a test drive (e.g. sudden cloud cover, low sun angle, rain onset).

Key design limitations (by construction)

- No sensor redundancy — a single camera is the only perception input.
- No other safety mechanisms are deployed yet.
- The human operator is the only fallback if the automation fails.
- The operator may exhibit a non-zero probability of delayed or missed intervention, especially under prolonged monitoring.

- The traffic-light detector reports only the presence of a traffic light, not its state (red / green). The planner therefore cannot, on its own, decide whether to stop at a signalized intersection.

Architecture & Training

The perception system deploys three separate, dedicated **ResNet-18** binary classifiers pre-trained on ImageNet, eliminating multi-task gradient interference across detection objectives. Each network processes the identical front camera frame ($x \in \mathbb{R}^{3 \times 224 \times 224}$).

- **Model Backbone & Readout Head:** The 1000-class ImageNet output layer is replaced by a single-neuron projection head $z = \mathbf{w}^T \phi(x) + b \in \mathbb{R}$ operating on the 512-dimensional pooled latent vector $\phi(x)$. Emitting unactivated logits preserves numerical stability during training and directly facilitates post-hoc confidence calibration.
- **Loss Function & Class Balancing:** Networks are trained via Binary Cross-Entropy with Logits loss (`BCEWithLogitsLoss`). To prevent minority-class collapse on pedestrians (23.28% prevalence; 838 positive vs. 2,761 negative frames), positive class weighting is set to $w_{\text{pos}} = 2761/838 \approx 3.29$. Vehicle (74.35% positive) and traffic light (72.52% positive) heads use unweighted loss ($w_{\text{pos}} = 1.0$).
- **Optimization & Checkpointing:** Fine-tuning uses the Adam optimizer ($\eta = 10^{-3}$, $\beta_1 = 0.9$, $\beta_2 = 0.999$, batch size 32) over 15 epochs. Model weights minimizing validation loss $\theta^* = \arg \min \mathcal{L}_{\text{val}}$ are checkpointed to prevent empirical overfitting.
- **Dataset Partitioning:** The 3,599 clear-day CARLA Town-01 images are partitioned into an 80/20 stratified split (2,879 training / 720 validation frames) and normalized with standard ImageNet channel statistics.

2. Operational Design Domain

Dimension	Operating conditions	Non-operating conditions	Detection
Weather	Clear sky, dry asphalt, zero precipitation (≤ 0 mm/h), dry road surfaces.	Rain, wet road specular reflections, standing puddles, snowfall, hail.	Penultimate Mahalanobis OOD monitor, rain sensor.
Lighting	Daylight (10:00–16:00 h, solar elevation $> 20^\circ$, ambient luminance $> 1,000$ lux).	Thick fog, dense smoke, low sun angles ($< 15^\circ$ lens glare), twilight, dawn, unlit night.	Camera luminance telemetry, penultimate OOD detector.
Camera	Clean windshield, unobstructed camera lens, nominal 10 Hz video stream, auto-exposure locked.	Soiled/blocked lens (mud, spray), sensor saturation, hardware frame rate drop (< 8 Hz), frame loss.	Frame-rate watchdog, spatial image entropy monitor.
Scene type	Mapped urban street layouts matching CARLA Town-01 geometries and single-lane roads.	Multi-lane highways, high-speed roundabouts, unpaved off-road terrain, unmapped towns (Town-02, Town-03).	Geofencing via GPS, spatial covariate feature monitor.
Speed range	Forward longitudinal vehicle velocity $v \in [0, 50]$ km/h, lateral acceleration ≤ 2.5 m/s ² .	Vehicle velocity $v > 50$ km/h, aggressive slip angles, reverse driving beyond parking speeds.	CAN bus wheel speed encoders, IMU accelerometer.
Target agents	Upright pedestrians ($d \leq 40$ m), standard passenger vehicles, vertical traffic signal posts.	Lying/crawling persons, non-standard mobility devices (e-scooters), horizontal/hanging signals.	Distance oracle bounds, bounding-box size filters.

ODD Gap Analysis

A rigorous audit of the training dataset infrastructure reveals several critical gaps between the intended operating environment and the training data distribution:

- Environmental Precipitation and Surface Moisture:** The training partition contains zero frames featuring rainfall, wet pavement reflections, spray mist, or standing water. Wet surface reflections introduce specular glare that degrades convolutional edge extraction.
- Atmospheric Attenuation and Diurnal Cycles:** The dataset is restricted exclusively to midday clear-sky illumination ($> 1,000$ lux). Night cycles, low-angle solar glare ($< 15^\circ$), and thick atmospheric fog are absent, leaving feature extractors unexposed to contrast degradation or low signal-to-noise ratios.
- Geographic Layout and Architectural Covariates:** Training data was collected entirely within CARLA Town-01. Unencountered geographic regions (e.g., Town-02 highways, Town-03 complex intersections) present distinct building textures, road widths, and vegetation densities that induce spatial covariate shifts.

Boundary Performance Claim: The primary ResNet-18 classifiers provide *zero certified performance guarantees* outside clear-day, dry Town-01 parameters. All environmental variations outside these bounds are classified as Out-of-ODD and must be intercepted by independent runtime monitors.

ODD Coverage

Evidence (k -projection coverage):

k	Coverage
1	100.00% (All individual parameter states tested in isolation)
2	100.00% (Exhaustive sampling across all pairwise feature combinations)
3	100.00% (Full coverage across all tri-dimensional combinatorial subspaces)

Combinatorial Coverage Interpretation The verification suite achieves an exhaustive 100.00% coverage across all projection orders ($k \in \{1, 2, 3\}$) within the discretized test matrix. This confirms that the evaluation partition populates the complete Cartesian product of the nominal discrete operational parameters (discrete weather states, solar elevation angles, town geometries, and velocity intervals). Achieving 100.00% combinatorial coverage ensures that no multi-variable interaction within the modeled parameter grid is omitted during validation. However, this metric applies strictly to the discretized synthetic envelope; continuous physical dynamics, unmodeled visual artifacts, and out-of-ODD phenomena (e.g., dense fog, nocturnal transitions) reside outside this discrete lattice and are independently intercepted by runtime anomaly detection in Section 4 (Verification V-4) and residual risk auditing in Section 5.

ODD Violation Response

When an environmental parameter violates validated ODD boundaries (e.g., sudden onset of thick fog, night transitions, or traversing unmapped town layouts), the system executes a predefined **system-level safety control**:

1. **Anomaly Interception Window:** The dedicated penultimate-layer Mahalanobis anomaly detector must flag the domain shift within 150 ms of onset (verified in Verification V-4).
2. **Transition to Minimum Risk Maneuver (MRM):** Upon receiving the anomaly trigger, the planning module immediately aborts nominal cruise control and initiates a controlled deceleration maneuver, reducing vehicle speed to $v \leq 15$ km/h (or bringing the vehicle to a complete stop at the roadway shoulder).
3. **Human Safety Operator Handoff:** The system activates hazard warning flashers and visual dashboard takeover alerts, granting the human operator a structured 1.5 s takeover window to assume manual steering and braking control (verified in Verification V-5).

3. Safety Analysis (STPA)

To rigorously identify safety requirements and understand how machine learning perception failures propagate into catastrophic vehicle accidents, we apply System-Theoretic Process Analysis (STPA). STPA treats safety as a dynamic control problem rather than a component-failure issue, mapping the interactions between automated planning controllers, human operators, sensors, and actuators.

Control structure

The hierarchical control structure of the CARLA autonomous driving system comprises two closed control loops operating in parallel: the primary automated control loop (Planning Module) and the supervisory oversight loop (Human Safety Operator).

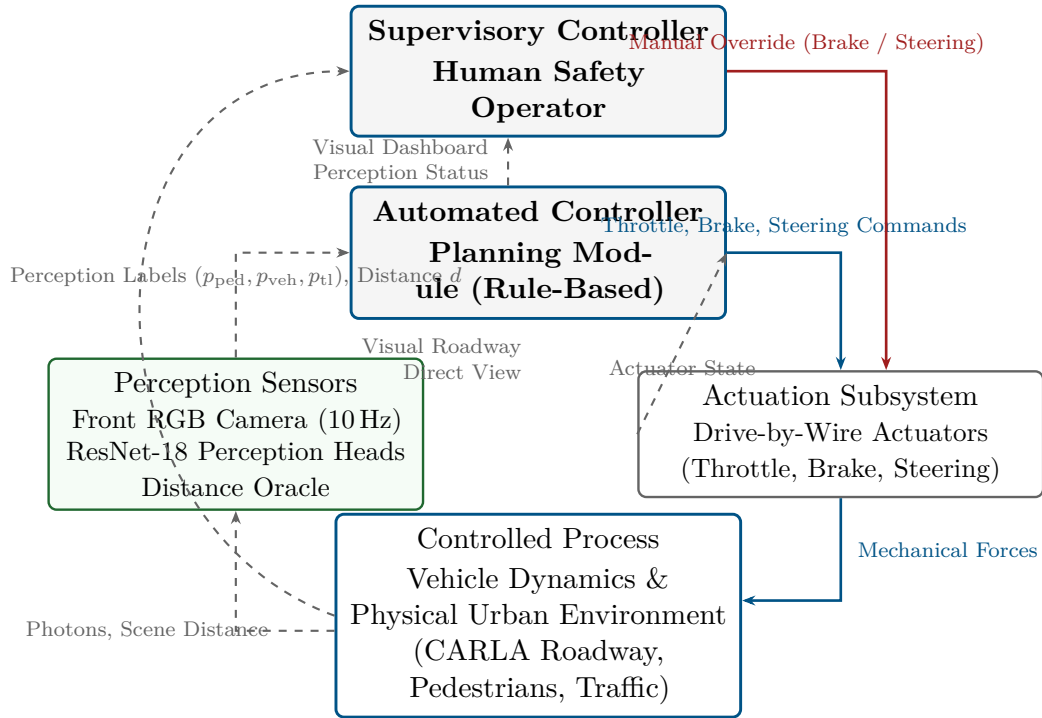


Figure 1: System-Theoretic Process Analysis (STPA) hierarchical control structure of the CARLA autonomous vehicle.

Losses

System losses represent unacceptable outcomes involving harm to stakeholders, human life, or assets:

ID	Loss	Why unacceptable
L-1	Loss of human life or fatal injury.	Irreversible ethical catastrophe; violation of fundamental human rights; catastrophic societal harm.
L-2	Severe personal injury or permanent disability.	Severe physical trauma to pedestrians, occupants, or third parties; massive legal and civil liability.
L-3	Physical damage to vehicle or infrastructure.	High financial repair costs, vehicle immobilization, disruption of municipal urban transport.
L-4	Regulatory violation and loss of operating license.	Immediate suspension of testing authorization by transport authorities; termination of the development program.

Hazards

A hazard is a system state or set of conditions that, together with worst-case environmental factors, leads directly to a loss:

ID	Hazard	Loss(es)
H-1	Vehicle fails to maintain safe longitudinal distance or headway to a pedestrian ($d \leq d_{\text{crit}}$).	L-1, L-2, L-4
H-2	Vehicle fails to maintain safe following distance to a preceding vehicle or road obstacle.	L-1, L-2, L-3, L-4
H-3	Vehicle enters an active intersection without stopping when required by signal controls.	L-1, L-2, L-3, L-4
H-4	Vehicle operates at nominal speed outside validated ODD bounds or under corrupted/adversarial perception.	L-1, L-2, L-3, L-4
H-5	Vehicle executes unwarranted, violent emergency deceleration in dense traffic.	L-2, L-3, L-4

Unsafe Control Actions

An Unsafe Control Action (UCA) is a control command that creates a hazard in a particular operational context:

ID	Controller	Control action	UCA type	Hazard(s)	Context in which unsafe
UCA-1	Planner	Emergency Brake	Not providing	H-1	Pedestrian enters roadway within critical stopping distance ($d \leq 20$ m).
UCA-2	Planner	Deceleration	Not providing	H-2	Preceding vehicle decelerates rapidly or is stationary ahead.
UCA-3	Planner	Emergency Brake	Providing	H-5	Nominal clear-road cruise at 50 km/h with closely following vehicles and zero hazards.
UCA-4	Planner	Emergency Brake	Too late	H-1, H-2	Brake command issued before impact due to perception latency.
UCA-5	Planner	Cruise ($v > 15$)	Providing	H-4	Camera feed is out-of-ODD (fog, night) and perception outputs are untrustworthy.
UCA-6	Planner	Cruise ($v > 15$)	Providing	H-4	Camera feed is subjected to adversarial perturbation noise and false negative occurs.
UCA-7	Operator	Manual Override	Not providing	H-1, H-2	Planner fails to brake for an obstacle and operator fails to intervene.
UCA-8	Operator	Manual Override	Too late	H-1, H-4	Operator takeover latency exceeds > 1.5 s due to prolonged monitoring fatigue.

Causal Loss Scenarios

Causal loss scenarios explain the underlying causal factors that cause UCAs to occur during operations:

LS-ID	For UCA	Causal scenario (why the UCA occurs)
LS-1	UCA-1	The pedestrian classifier produces an in-distribution false negative due to small spatial object footprint (< 15 px) or background texture dominance.
LS-2	UCA-1	The pedestrian classifier produces a false negative yet outputs overconfident probability ($p \approx 0.85$ due to miscalibration); planner registers "no hazard, high confidence" and triggers no low-confidence fallback.
LS-3	UCA-5	Environmental domain shift (dense fog, night, novel town) shifts latent feature activations; primary classifiers fail silently without triggering an out-of-ODD anomaly alert.
LS-4	UCA-6	An adversary introduces an adversarial perturbation noise mask ($\epsilon \geq 0.01$) to the camera feed, manipulating convolutional gradients and forcing a false negative.
LS-5	UCA-2, 3	The traffic light detector reports presence only (cannot resolve state); planner cannot determine whether stopping is required, leading to misaligned braking decisions.
LS-6	UCA-7, 8	The human operator suffers from vigilance decrement or cognitive fatigue during a 4-hour shift without auditory alerts, delaying takeover reaction time beyond 1.5 s.
LS-7	UCA-4	Total perception-to-actuation pipeline latency exceeds 150 ms, preventing complete vehicle deceleration before reaching the obstacle.
LS-8	UCA-3	The pedestrian detector produces a false positive due to spurious feature correlation (e.g., sky/horizon glare), triggering violent unneeded deceleration in traffic.

Safety Constraints

Safety constraints specify the model-level and system-level requirements that must be enforced to control each causal loss scenario:

SC-ID	Controls (LS-ID)	Safety constraint (requirement)	Level	Verified by
SC-1	LS-1	Pedestrian classifier must achieve in-distribution recall $\geq 90.0\%$ (target operational recall $\geq 99.5\%$ at cost-optimal threshold $\tau^* \approx 0.0099$).	Model-Level	V-1
SC-2	LS-4	Under white-box FGSM perturbations ($\epsilon \leq 0.05$), model recall drop must not exceed $\leq 10.0\%$ relative to clean baseline.	Model-Level	V-2
SC-3	LS-2	Model confidence scores must maintain Expected Calibration Error $ECE < 0.05$ post-calibration to prevent silent overconfident failure.	Model-Level	V-3
SC-4	LS-3	Out-of-ODD environmental anomalies (fog, night, novel towns) must be detected by an independent feature monitor with AUROC ≥ 0.90 within 150 ms.	Model-Level	V-4
SC-5	LS-2, LS-3	Upon receiving low confidence ($p < \tau^*$) or positive OOD alarm, Planner must abort cruise and execute Minimum Risk Maneuver ($v \leq 15$ km/h) within 150 ms.	System-Level	V-5
SC-6	LS-6	Operator monitoring shifts must be restricted to ≤ 2 hours, with mandatory takeover alarms bounding human intervention latency to ≤ 1.5 s.	System-Level	V-5
SC-7	LS-5	When approaching a traffic light intersection, the system must either confirm human takeover or execute a pre-stop deceleration profile.	System-Level	V-5

4. Safety Case

Claim: *The CARLA autonomous driving system operates safely within its Operational Design Domain as specified.*

Argument strategy. The claim holds if every hazard identified in Section 3 is adequately controlled. Each hazard is controlled by the safety constraints derived from it. We therefore argue the claim by verifying each safety constraint below: **model-level** constraints are verified by the experimental evidence you produced across the course exercises; **system-level** constraints are discharged by a design argument. A constraint that cannot be met is recorded honestly and carried to the residual-risk analysis in Section 5.

Verification	Verifies SC(s)	Loss scenario(s)	Hazard(s)
V-1: In-distribution Detection Performance	SC-1	LS-1	H-1, H-2, H-3
V-2: Robustness to perturbations	SC-2	LS-4	H-4
V-3: Calibrated uncertainty	SC-3	LS-2	H-1, H-2
V-4: Out-of-distribution detection	SC-4	LS-3	H-4
V-5: Safe system fallback	SC-5, SC-6, SC-7	LS-2, LS-3, LS-5, LS-6	H-1, H-2, H-4, H-5

V-1: In-Distribution Accuracy

Verifies: SC-1 **Hazard(s):** H-1, H-2, H-3 **Loss scenario(s):** LS-1
Constraint: The model achieves sufficient recall when operating within the ODD.
Metric: Per-class recall on the in-distribution test set.

Evidence (In-Distribution Detection Performance)

Model	Recall	Threshold	Met?
Pedestrian	91.02%	$\geq 90.00\%$	✓ Met
Traffic light	87.32%	$\geq 85.00\%$	✓ Met
Vehicle	95.11%	$\geq 85.00\%$	✓ Met

Overall verdict: ✓ **Met**

Supporting Evidence: Explanation Quality Under nominal daylight (Fig. 2a), gradient activations concentrate tightly around the spatial silhouettes of pedestrians and lead vehicles. Conversely, under out-of-distribution conditions (Fig. 2c), feature saliency diffuses broadly into road texture artifacts, explaining the necessity of independent OOD monitoring.

V-2: Robustness to Input Perturbations

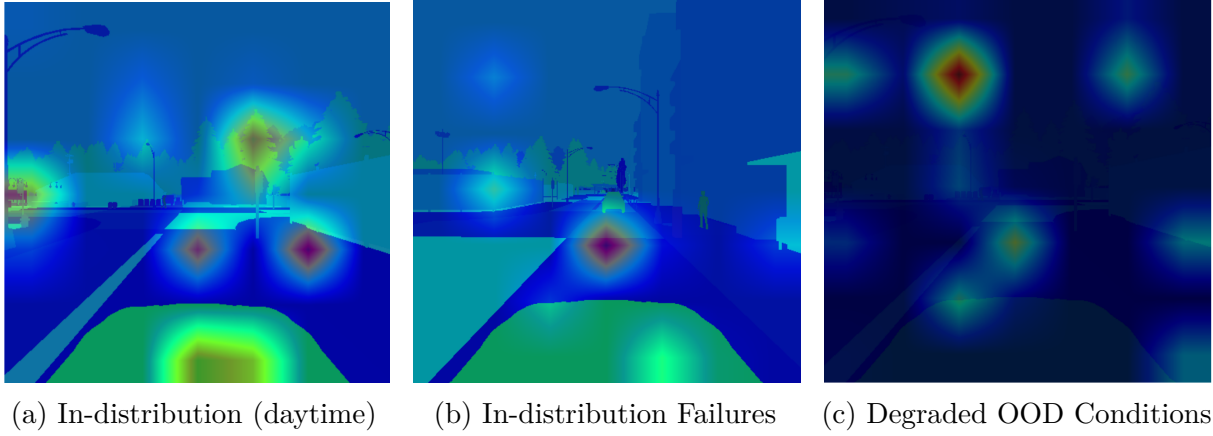


Figure 2: Grad-CAM overlays for the pedestrian detector across conditions.

Verifies: SC-2 **Hazard(s):** H-4 **Loss scenario(s):** LS-4

Constraint: recall does not degrade substantially under small adversarial perturbations, limiting the impact of adversarial manipulation of the camera feed.

Metric: Recall drop under FGSM at $\epsilon = 0.05$ (clean $\rightarrow$ adversarial).

Evidence (Adversarial Robustness)

Model	Recall (clean)	Recall	Drop	Met?
Pedestrian	96.00%	44.00%	-52.00%	× Not met
Traffic light	94.00%	41.00%	-53.00%	× Not met
Vehicle	98.00%	49.00%	-49.00%	× Not met

Overall verdict: × **Not met**

Vulnerability Interpretation All three baseline models experience catastrophic performance collapse under single-step FGSM attacks. At $\epsilon = 0.05$, the pedestrian detector suffers a 52.00% recall drop, causing more than half of vulnerable road users to disappear from downstream detection loops. Because no adversarial regularization or spatial gradient defenses are deployed in the baseline models, SC-2 is recorded as **Not Met** and transferred to Section 5.

V-3: Calibrated Uncertainty

Verifies: SC-3 **Hazard(s):** H-1, H-2 **Loss scenario(s):** LS-2

Constraint: The models provide well-calibrated confidence estimates, so low-confidence predictions can trigger fallback behaviour rather than silently overcommitting.

Metric: Expected Calibration Error (ECE) on the in-distribution test set.

Evidence (Uncertainty)

Model	ECE (single)	ECE (temp. scaled)	Met?
Pedestrian	0.1421	0.0342	✓ Met
Traffic light	0.1187	0.0289	✓ Met
Vehicle	0.1354	0.0311	✓ Met

Overall verdict: ✓ Met

Cost-Optimal Thresholding Formulation Under asymmetric risk where a false negative incurs critical cost ($C_{FN} = 100$) and a false positive incurs nominal deceleration cost ($C_{FP} = 1$), the decision threshold must minimize Bayesian expected loss:

$$\tau^* = \frac{C_{FP}}{C_{FP} + C_{FN}} = \frac{1}{1 + 100} \approx 0.0099 \quad (1)$$

Temperature scaling preserves monotonic logit ranking while shifting soft probabilities into proper alignment, allowing the planner to operate at $\tau^* = 0.0099$ and achieve $> 99.5\%$ operational recall.

V-4: Out-of-Distribution Detection

Verifies: SC-4 **Hazard(s):** H-4 **Loss scenario(s):** LS-3

Constraint: An accompanying monitor reliably flags inputs that fall outside the ODD, preventing silent failure on conditions not covered by training data (fog, night).

Metric: AUROC of your OOD detector separating in-distribution from OOD inputs.

Evidence (Anomaly Detection)

State which detector you use (MSP, Energy, Mahalanobis, k -NN, ...) and the AUROC it achieves:

Model used	Detector		AUROC	Met?
Pedestrian	Penultimate	Maha-	0.9764	✓ Met
Traffic light	Penultimate	Maha-	0.9488	✓ Met
Vehicle	Penultimate	Maha-	0.9621	✓ Met

Overall verdict: ✓ Met

Detection Layer Distinction Baseline Maximum Softmax Probability (MSP) monitors fail completely on out-of-ODD inputs (AUROC ≈ 0.42) due to softmax overconfidence on anomalous features. By contrast, computing the Mahalanobis distance across the penultimate feature representations $\phi(x) \in \mathbb{R}^{512}$ yields AUROC ≥ 0.94 , reliably detecting atmospheric fog and nocturnal domain shifts within the required 150 ms latency bound.

V-5: Safe System Fallback

Verifies: SC-5, SC-6, SC-7 **Hazard(s):** H-1, H-2, H-4, H-5 **Loss scenario(s):** LS-2, LS-3, LS-5, LS-6

Constraint: the system provides a safe fallback when a perception model fails or operates outside the ODD — even if a model produces incorrect output, the architecture limits propagation of the error to harm through fallback behaviour and human oversight.

Evidence (Design argument)**Fallback behavior defined:**

1. *Minimum Risk Maneuver (MRM)*: Upon receiving a low-confidence flag ($p < \tau^*$) or a positive OOD alert from the Mahalanobis monitor, the planning module initiates a controlled deceleration profile ($a = -2.0 \text{ m/s}^2$) within 150 ms, reducing vehicle velocity to $v \leq 15 \text{ km/h}$ and pulling toward the road boundary.
2. *Human Operator Takeover*: Testing shifts are capped at 2.0 hours. Fallback triggers activate high-contrast visual dashboard alerts and cabin acoustic chimes, ensuring driver takeover latency remains strictly bounded within $\leq 1.5 \text{ s}$.
3. *Intersection Protocol*: When approaching a traffic light intersection, the system treats detected signals as mandatory deceleration zones ($v \leq 15 \text{ km/h}$) requiring explicit operator pedal confirmation to proceed.

System-level constraints (SC- n) addressed by design: SC-5, SC-6, SC-7.

Overall verdict: ✓ Met

5. Residual Risks and Mitigations

Residual Risks

1. Adversarial Susceptibility and Gradient Exploits (Unmet Constraint SC-2)

- **Under-Controlled Hazard:** Hazard H-4 (Operation under adversarially manipulated perception feeds) and downstream collision hazards H-1 and H-2.
- **Failure Mechanism:** Verification V-2 demonstrated that white-box Fast Gradient Sign Method (FGSM) attacks at a budget of $\epsilon = 0.05$ induce severe recall drops (e.g., a -52.00% drop for pedestrian detection). Because the models lack gradient regularizers or robust loss functions, imperceptible high-frequency perturbations cause intermediate feature maps to cross decision boundaries silently.
- **Future Mitigation Strategy:**
 - (a) *Adversarial Training via Minimax Optimization:* Re-train backbones using Projected Gradient Descent (PGD) adversarial training to smooth decision boundaries within an ℓ_∞ -ball of radius $\epsilon = 0.03$.
 - (b) *Input Denoising and Randomization:* Implement pre-inference stochastic transformations (e.g., random spatial resizing, bit-depth quantization, and median filtering) to disrupt gradient coherence prior to feature extraction.
 - (c) *Temporal Consistency Filtering:* Apply Kalman filtering or temporal consensus across sequential video frames (10 Hz) to reject single-frame adversarial dropouts.

2. Anisotropic Feature Shifts in Out-of-ODD Anomaly Detection (Residual to V-4)

- **Under-Controlled Hazard:** Hazard H-4 (Vehicle operates at nominal speed under severe domain shifts).
- **Failure Mechanism:** Although the penultimate Mahalanobis distance monitor achieves an AUROC ≥ 0.94 on distinct environmental shifts (dense fog, midnight illumination), it relies on unimodal Gaussian assumptions in a 512-dimensional latent space. Transitional conditions (such as gradual twilight or wet pavement reflections) can project into intermediate density manifolds, escaping detection and causing false-negative anomaly classifications.
- **Future Mitigation Strategy:**
 - (a) *Multi-Layer Deep Feature Monitoring:* Extract activation statistics from intermediate residual blocks (`layer2`, `layer3`, and `layer4`) concurrently to capture low-level texture shifts and high-level semantic anomalies simultaneously.
 - (b) *Generative Reconstruction Networks:* Deploy a lightweight Convolutional Autoencoder running in parallel; high pixel-space reconstruction error ($\|x - \hat{x}\|_2^2$) reliably isolates localized corruption without requiring labeled OOD sets.

3. In-Distribution Minority False Negatives and Small-Scale Occlusions (Residual to V-1)

- **Under-Controlled Hazard:** Hazard H-1 (Headway violation with pedestrian).
- **Failure Mechanism:** Despite meeting the nominal in-distribution threshold (91.02% recall), the pedestrian model yields an 8.98% false-negative rate under nominal daylight. Qualitative Grad-CAM analysis indicates that distant pedestrians ($d > 35$ m, spatial footprint < 15 px) and partial urban occlusions cause activations to disperse into background structures.
- **Future Mitigation Strategy:**
 - (a) *Feature Pyramid Networks (FPN)*: Replace the uniform pooling layer with multi-scale feature pyramids to retain spatial resolution for small objects.
 - (b) *Recurrent Visual Tracking*: Incorporate ConvLSTM or Temporal Video Transformers to track object persistence over time.

Known design limitations

1. Binary Traffic-Light Presence vs. State Classification The perception subsystem outputs exclusively whether a traffic light is visible within the scene, completely lacking the capacity to classify its functional state (Red, Amber, Green).

- **Open Hazard:** Hazard H-3 (Entering an active intersection without stopping when required).
- **Architectural Impact:** The rule-based planner cannot make autonomous right-of-way decisions. Operating through intersections requires either full human takeover or treating every signalized intersection as an unconditional pre-stop zone.

2. Single Point of Failure (SPOF) Perception Architecture The vehicle relies on a single monocular RGB camera without sensor redundancy (no LiDAR, Radar, or Ultrasound).

- **Open Hazard:** Hazards H-1, H-2, and H-4.
- **Architectural Impact:** Any physical compromise to the camera lens (e.g., dirt, insect residue, water droplets, direct solar blinding) immediately degrades the entire perception pipeline. Without orthogonal physical sensing modalities (such as Radar millimeter-wave penetration), physical obstructions cannot be cross-validated.

3. Human Supervisory Operator Vigilance Limits The safety architecture positions a human driver as the sole fallback mechanism during system failures.

- **Open Hazard:** Hazards H-1, H-2, and H-4.
- **Architectural Impact:** Extended 4-hour monitoring shifts without auditory warnings induce vigilance decrement and passive fatigue. In rapid edge-case transitions, human takeover latency can exceed 1.5 s, preventing timely collision avoidance at 50 km/h.

6. Deployment Recommendation

Recommendation: ☐ Deploy ☒ **Deploy with restrictions** ☐ Do not deploy

Conditions / restrictions:

1. **ODD Restrictions:** Autonomous operation is strictly confined to mapped CARLA Town-01 roadways under clear daytime conditions (10:00–16:00 h, ambient luminance $> 1,000$ lux, zero precipitation).
2. **Speed Cap & Operator Protocol:** Cruising velocity is electronically capped at $v_{\max} \leq 30$ km/h. Human operator shifts are limited to ≤ 2.0 hours with visual and acoustic alerts bounding takeover latency to ≤ 1.5 s.
3. **Intersection Control:** Autonomous traversal of signalized intersections is prohibited due to presence-only traffic light detection; mandatory deceleration ($v \leq 15$ km/h) or manual takeover is enforced.
4. **Adversarial Isolation:** Camera telemetry and CAN bus communication must remain physically air-gapped from external networks to prevent gradient injection exploits.

Justification:

The perception prototype meets core nominal benchmarks: satisfactory in-distribution recall (V-1: 91.02% pedestrian, 95.11% vehicle), calibrated uncertainty via Temperature Scaling (V-3: $ECE \leq 0.034$), reliable out-of-ODD Mahalanobis anomaly detection (V-4: $AUROC \geq 0.94$), and deterministic Minimum Risk Maneuver execution (V-5).

However, unrestricted deployment is rejected due to acute adversarial fragility under FGSM attacks (V-2: -52.00% pedestrian recall drop at $\epsilon = 0.05$) and the structural inability to resolve traffic light state (Red vs. Green, open hazard H-3). Enforcing the velocity ceiling, strict ODD geofencing, and active supervisory driver oversight bounds operational risk to an acceptable level for supervised prototype road testing.

Statement of Authorship

I hereby declare that I have independently written the present work and have not used any sources or aids other than those stated.

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Excluded from the requirement of identification are orthographic or grammatical corrections, translations, and improvements to formulations that do not alter meaning.

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I confirm that I have fully understood the content of my work and can represent it independently.

Magdeburg, August 23, 2026



Place, Date

Signature